CHAPTER 18

Prototyping

Highlights

- Prototyping intertwines with other UX design activities
- Depth and breadth of prototypes
  - Horizontal prototypes
  - Vertical prototypes
  - Local prototypes
  - T prototypes
- Wireframe prototypes
- General process of representing interaction
- Fidelity of prototypes
- Building up prototypes in increasing levels of fidelity
- Click-through prototypes
- Specialized prototypes

18.1 INTRODUCTION

18.1.1 You Are Here

We begin each process chapter with a "You are here" picture of the chapter topic in the context of The Wheel, the overall UX design lifecycle template (Fig. 18.1). In this chapter, we describe how to perform the prototype candidates UX design lifecycle activity.

In the design chapters of Part 3, especially Chapter 13 on generative design, we discussed various techniques for ideation to generate large number of design ideas for exploration. We captured those ideas using sketches (fast and low-fidelity prototypes) and subjected them to analysis through critiquing to identify promising design ideas. Here, prototyping is a full-fledged lifecycle activity to realize, envision, and evaluate those promising design candidates by expanding on their scope, detail, and understanding. In this chapter, we describe types of prototypes and, for interaction design, how to make wireframe prototypes. As designs evolve, designers produce various kinds of

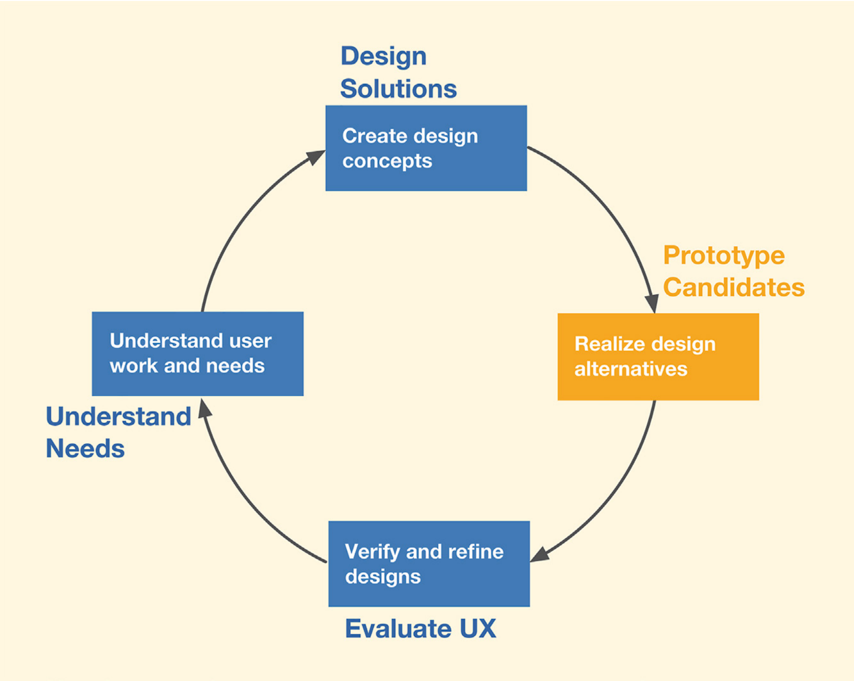

Fig. 18.1

You are here in the chapter on the prototype candidates activity in the context of the overall Wheel UX lifecycle.

prototypes as external design representations. Because prototypes are made for many different purposes, there are different kinds of prototypes, each with unique capabilities and tradeoffs.

18.1.2 Prototyping Intertwines With Other UX Activities

Recall in Chapter 5 we said each lifecycle activity will be described separately in its own chapter(s), but in reality they all go together and occur in combination; they are closely intertwined and interleaved throughout the lifecycle.

Prototyping is a good example of this intertwining. Prototyping occurs at the start of design creation in the form of sketches and continues to occur as wireframes and other forms throughout much of the remaining design process.

18.1.3 A Dilemma and a Solution

The only way to be sure that your design is the right design and the best design it can be is to subject it to UX evaluation. However, at some early point, you will have a design but no instantiation of it as a product or system yet to evaluate. If you wait until after it is implemented, however, changes are much more difficult and expensive to make. A prototype gives you something to evaluate before you have to commit resources to build the real thing. Prototypes allow you to fail faster, learn sooner, and succeed earlier.

18.1.4 Advantages of Prototyping

Prototypes have the following advantages:

- Provide a platform to support UX evaluation with users
- Offer concrete baseline for communication between users and designers
- Provide a conversational prop to support communication of concepts not easily conveyed verbally
- Allow users to try the design
- Give project visibility and buy-in within customer and developer organizations
- Encourage early user participation and involvement
- Give the impression that some design changes are easy to make, so now is the time to make them, if needed
- Afford designers immediate observation of user performance and consequences of design decisions
- Help sell management on an idea for new product
- Help affect a paradigm shift from an existing system to a new system

18.1.5 Universality of Prototyping

The idea of prototyping is timeless and universal. Automobile designers build and test mockups, architects and sculptors make models, circuit designers use breadboards, artists work with sketches, and aircraft designers build and fly experimental designs. Even Leonardo da Vinci and Alexander Graham Bell made prototypes.

Thomas Edison is famous for making thousands of prototypes before getting just the right design. In each case, the concept of a prototype was key to affording the design team and others an early ability to observe something about the final product—evaluating ideas, weighing alternatives, and seeing what works and what does not.

Alfred Hitchcock, master of dramatic dialogue design, is known for using prototyping to refine the plots of his movies. Hitchcock would tell variations of stories at cocktail parties and observe the reactions of his listeners. He would experiment with various sequences and mechanisms for revealing the storyline. Refinement of the story was based on listener reactions as an evaluation criterion. The movie *Psycho* is a notable example of the results of this technique.

See Section 18.8 for Scandinavian origins of prototyping.

18.2 DEPTH AND BREADTH OF A PROTOTYPE

The idea behind prototypes is to provide fast and easily changed early views of an envisioned UX design. Because it must be quickly and easily changed, a prototype is a design representation that is in some way(s) less than a full implementation. The choices for your approach to prototyping are about how to make it less. One way you can make it less is by focusing on just the breadth or just the depth of the system.

When you slice the features and functionality of a system by breadth, you get a horizontal prototype. When you slice by depth, you get a vertical prototype (Hartson & Smith, 1991). In *Usability Engineering*, Nielsen (1993) illustrates the relative concepts of horizontal and vertical prototyping (Fig. 18.2).

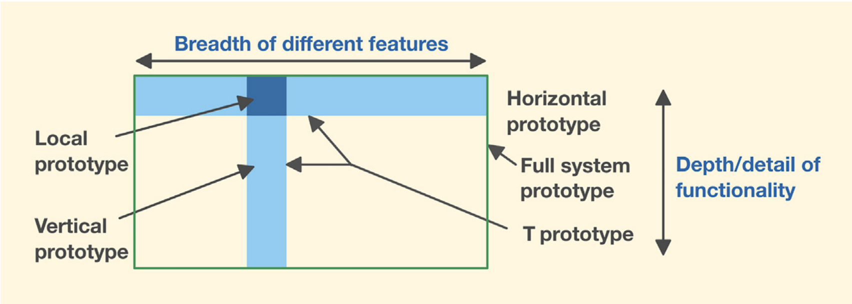

Fig. 18.2
Horizontal and vertical prototyping concepts (adapted from Nielsen, J. (1993). Usability engineering. *Courtesy of Academic Press Professional.).*

18.2.1 Horizontal Prototypes

A horizontal prototype (top bar in Fig. 18.2) is very broad in the features it incorporates, but it offers less depth in its coverage of how that functionality works. A horizontal prototype is a good place to begin with your prototyping, as it provides an overview on which you can base a top-down approach. Prototyping in the early funnel tends to be horizontal in nature. The early part of the agile UX funnel model is for large-scope activity, usually for conceptual design, before syncing with software engineering (SE) (see Section 4.4.4).

A horizontal prototype is effective in demonstrating and validating the product concept and for conveying an early product overview to managers, customers, and users (Kensing & Munk-Madsen, 1993). However, because of the lack of details in depth, horizontal prototypes usually do not support complete workflows. As such, task-based user experience evaluation with this kind of prototype is generally less realistic. They are great for user experience evaluation focused on concept validation and ensuring the emerging early designs align with the broader work goals of the users.

18.2.2 Vertical Prototypes

A vertical prototype (upright bar in Fig. 18.2) contains more depth of detail for some functionality, but only for a narrow selection of features. A vertical prototype allows evaluation of a limited range of features, but those parts that are included have evolved in enough detail to support realistic user experience evaluation. Often the functionality of a vertical prototype can include a stub for or a connection to an actual working backend database.

A vertical prototype is ideal for times when you need to represent completely the details of an isolated part of an individual interaction workflow to understand how those details play out in actual usage. For example, you may wish to study a new design for the checkout part of the workflow for an ecommerce website. A vertical prototype may show in depth that one task sequence and associated user actions. Because vertical prototypes are usually about individual features, they are most commonly used in the late funnel part of the process. The late funnel is the part of the agile UX funnel model for small-scope activity and for syncing with agile SE sprints (see Section 4.4.3).

18.2.3 T Prototypes

The T prototype combines advantages of both the horizontal and vertical prototypes (T-shaped area in Fig. 18.2), offering a good compromise for design evaluation. Much of the feature breadth is realized at a shallow level (the top of the T), but a few key parts are done in depth (the vertical part of the T).

In the early going, we recommend the T prototype because it provides a nice balance between the two extremes, giving you some advantages of each. Once you have established a system overview in your horizontal prototype, as a practical matter the T prototype is the next step toward achieving some depth. In time, the horizontal foundation supports evolving vertical growth across the whole prototype.

18.2.4 Local Prototypes

Sometimes you need a local prototype, which is narrow in both dimensions, limiting its focus to a localized interaction design issue. A local prototype is used to evaluate design alternatives for particular isolated interaction details, such as one dialogue box, the appearance of an icon, the wording of a message, or the behavior of an individual interaction object.

A local prototype is the solution for those times when your design team encounters an impasse in design discussions where, after a while, there is no agreement, and people are starting to repeat themselves. Perhaps UX research data are not clear on the question, and further arguing is a waste of time. It is time to put the specific design issue on a list for evaluation, letting the user or customer speak to it in a kind of feature face-off to help decide among the alternatives.

Because of their role in deciding specific issues, local prototypes are used independently from other prototypes and are temporary and disposable, having short life spans.

18.2.5 Examples of Kinds of Prototypes

To illustrate these kinds of prototypes, we use an overly simple example so the details of the device design do not distract from prototyping concepts. We make no claim that any of these prototypes are good, only that they are used as illustrations.

Consider the design of a digital alarm clock app. Possible user tasks include set time, set alarm, set date, lock in settings, and initiate snooze cycle.

18.2.5.1 Example of horizontal prototype

In a horizontal prototype, you show all the buttons and other operable interaction objects on the home or main landing page of the app. Clicking on any button will lead to the first step in the prototype for the corresponding task, but no further details about doing any task are available. For example, clicking on the "Set-time" button shows the first screen of the task for setting the current time into the clock. Further clicking will not lead to further task details, however. No tasks are represented in detail, but stubbed for the moment.

This is a horizontal prototype because its scope is visualized as a horizontal bar across the tops of all the tasks available from the main landing page of the app (top bar in Fig. 18.2). You would use a horizontal prototype like this early in design discussions when you want to see the full breadth of task possibilities.

18.2.5.2 Example of vertical prototype

In a vertical prototype, clicking on any button except a selected one, say the "Set-time" button, shows the existence of the corresponding task/function but no detail. Clicking on the selected button corresponding to the task whose details are fleshed out (the "Set-time" task in this example) does take the user deeper into all the details of the "Set-time" task sequence.

This is a vertical prototype because its scope is visualized as a vertical path down through the details of one task (vertical bar of Fig. 18.2). You would use this kind of prototype to explore the design possibilities for a single feature or function.

18.2.5.3 Example of T prototype

In a T prototype, all first-level buttons take you only one level deep, except the selected task, which is seen in full detail. In this example, the "Set-time" button leads to the details of the specific task.

This is a T prototype because, as a combination of the horizontal and vertical prototypes, its scope is visualized as a horizontal bar plus a vertical bar (T-shaped area of Fig. 18.2). You would use a T prototype like this early in an agile environment when you want to develop only one feature/task at a time.

18.2.5.4 Example of local prototype

A local prototype could be used to explore alternatives for one point of interaction within one task/function. For example, in the task for setting the date, consider multiple ways of entering a date: (1) up and down arrows to set values for day of week, month, date, year; (2) spinners for the same; and (3) a popup calendar using left and right arrows to navigate and select a date on the calendar to enter that date into the clock.

This is a local prototype because it focuses on a very small (local) interaction in the design. The scope is one interaction within one task, visualized as a small area of intersection between a horizontal prototype and a vertical prototype (dark blue square in Fig. 18.2). You would use this kind of prototype to explore alternatives for design of an instance of interaction within a single feature or function.

18.3 WIREFRAME PROTOTYPES

Wireframes, the go-to technique for interaction design prototypes, were introduced in Section 16.6, where we described them as two-dimensional sketches or drawings consisting of shapes, connecting lines, and some text for labels (Fig. 18.3). Each shape represents a single interaction object, pattern, page, or screen (in the broadest sense of screen). The bulk of the wireframes will be made during designing for interaction (see Chapter 16).

When wireframes are connected together to represent possible navigational paths, that network of wireframes and connections is called a wireframe prototype (see Fig. 18.3), which is alternatively called a wireflow (Laubheimer, 2016) or wireflow prototype. We will stick with the simpler and more popular use of wireframe prototype.

A wireframe prototype is, at its base, a user workflow diagram (see Fig. 18.3) of the following:

- Boxes, representing system states, each of which is a wireframe, possibly representing a screen on a device
- Arrows representing navigational flow among the wireframes (transitions due to user interaction)

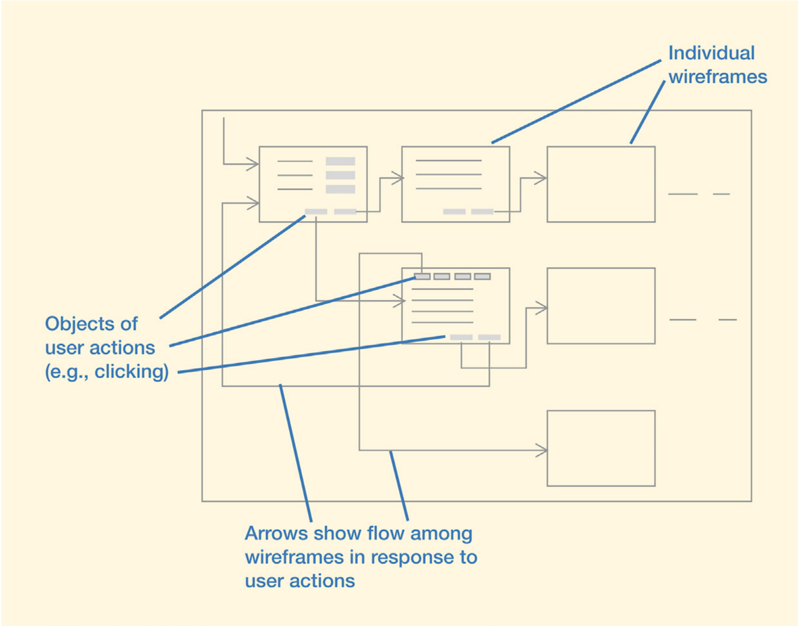

Fig. 18.3
Generic wireframe prototype.

Notice in Fig. 18.3 that a flow arrow goes from an interaction object (such as a button, icon, or menu choice) within one wireframe navigating to another wireframe. The process of establishing a wireframe interaction design prototype to represent an interaction design for a set of related tasks is summarized in Fig. 18.4.

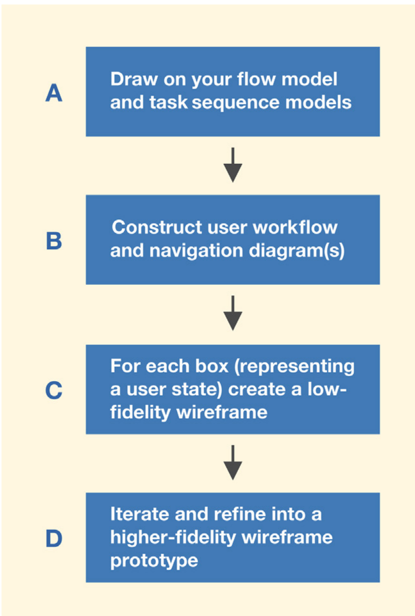

Fig. 18.4
The general evolutionary process of going from UX research models to wireframe prototypes.

18.3.1 Draw on Everything You Have Created for the Prototype Design

Use your conceptual design, design scenarios, ideation, personas, storyboards, and everything else you have created in working up to this first real materialization of your design ideas. Draw on your flow model and task-sequence models to represent flow and navigation (box A in Fig. 18.4).

18.3.2 Represent Flow and Navigation

Using your task-sequencing models, the next step in prototyping task sequencing and navigation is the creation of one or more user flow diagrams (box B in Fig. 18.4), which takes us a step closer to design by helping us represent details of flow and navigation in the interaction view of design. Begin with the main navigational paths, the essence of the flow, and initially leave out unnecessary detail, special conditions, and edge cases, such as error checking and confirmation dialogue.

Example: User Flow Diagram for a Form-Based User Task to Bundle Network Services[1]

The Network Infrastructure and Services group, part of the IT organization at Virginia Tech, is responsible for providing network and communications services to faculty, staff, and students. They maintain systems for ordering, billing, and maintaining services, such as wireless internet access for buildings and classrooms and wired ethernet for faculty offices and laboratories. Services also include all campus telephones and cable television for classrooms.

As part of their mission, hardware, software, and UX people develop systems to support customers in ordering network services. This example is about one small capability in that large portfolio of applications: a web-based feature for bundling services together in popular configurations. To understand this example, you need to know that each service can be offered in a plan (e.g., a choice of ethernet connection speed), and each plan can have add-ons, which are additional related features. A specialist in the Network Infrastructure and Services group is authorized to create these bundles, which can then be ordered by customers.

The system feature that supports bundle creation was the target of agile UX design. Beginning with raw UX research data from interviews with these specialists, we created a flow model and several task-sequence models. These

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[1]Thanks for permission to use this example to Joe Hutson and Mathew Mathai, Network Infrastructure and Services, Virginia Tech.

were used to create an early user flow diagram (Fig. 18.5) to illustrate the main sequencing of workflow for this user task.

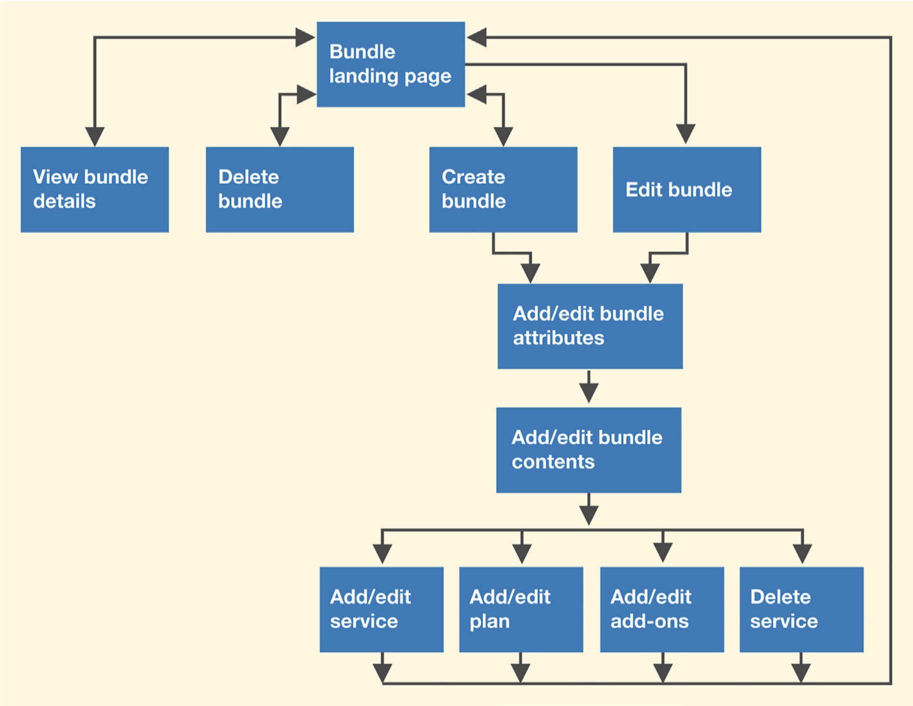

Fig. 18.5

Early user flow diagram for service bundling (courtesy of Joe Hutson and Mathew Mathai, Network Infrastructure and Services, Virginia Tech.).

Beginning at the "Bundle landing page," users can view, delete, or edit an existing bundle or create a new bundle. Editing a bundle includes editing bundle attributes (e.g., name, cost) and editing bundle contents (e.g., services, plans, and add-ons in the bundle). Once those changes are saved, the user is taken back to the "Bundle landing page." It looks straightforward in this diagram, which could be a testimony to the power of a good user flow diagram because it took quite a bit of analysis and synthesis of the disparate pieces of UX research data to get to this tidy representation.

In early design, your user flow diagrams can be translated almost directly into the structure of a wireframe deck (Section 18.3.1).

18.3.3 Create a Wireframe for Each State

Each state (box) of the user flow diagram becomes a low-fidelity wireframe design for one screen (box C in Fig. 18.4). Within that screen, you are designing for things that live in that state, such as the workspaces and dialogue to support the related task, including such elements as buttons, icons, pull-down menus, and forms to be filled in by the user. The design of wireframes is detailed in Section 16.6, and Fig. 16.8 contains an example of a wireframe design.

Arcs of the user flow diagram will guide you in adding the controls (e.g., buttons) to support navigation among the wireframes. As you work, you will probably have to add new states and the corresponding navigation to handle non-mainstream dialogue (e.g., to handle confirmation requests, such as “Are you sure you want to delete this service?”).

18.3.4 Iterate and Refine the Wireframe Prototype

After you have fleshed out the details of the individual screens, demonstrate the prototype to stakeholders, especially clients, to get feedback to iterate and refine the design (box D in Fig. 18.4). Your result will be a high-fidelity version with the wireframes in a navigational context (Fig. 18.6). Space limitations prevent showing this large enough to read the text, but you get the idea.

18.3.5 Wireframe Decks

When you break down a wireframe prototype into a set of individual wireframes, all related to the same task, you can group them as slides in a slide show. If you think of the individual wireframes as cards in a deck, it is called a wireframe deck. You can navigate through a wireframe deck as one would go through a deck of presentation slides, each slide being a screen. This allows you to flip from one state (wireframe screen) to another as you simulate transitions due to user inputs.

Each stakeholder involved will have a copy of this wireframe prototype on their laptop for design discussions in our UX studio.

18.3.6 Use Your Design System as a Library of Interaction Object Templates in Your Sketching Tool

You can add efficiency as you build up these wireframes through tools with libraries of interaction and user interface styles and objects (icons, buttons, menu bars, pull-down menus, pop-ups, etc.). Do not waste time by repeatedly building these interaction objects. You can achieve higher fidelity with minimal costs by fashioning these objects to match the style specifications in your design system (see Section 16.9), a kind of design library.

You can set yourself up with a design system containing particular styles (e.g., Windows or Mac styles or your own branded style). If you do not have

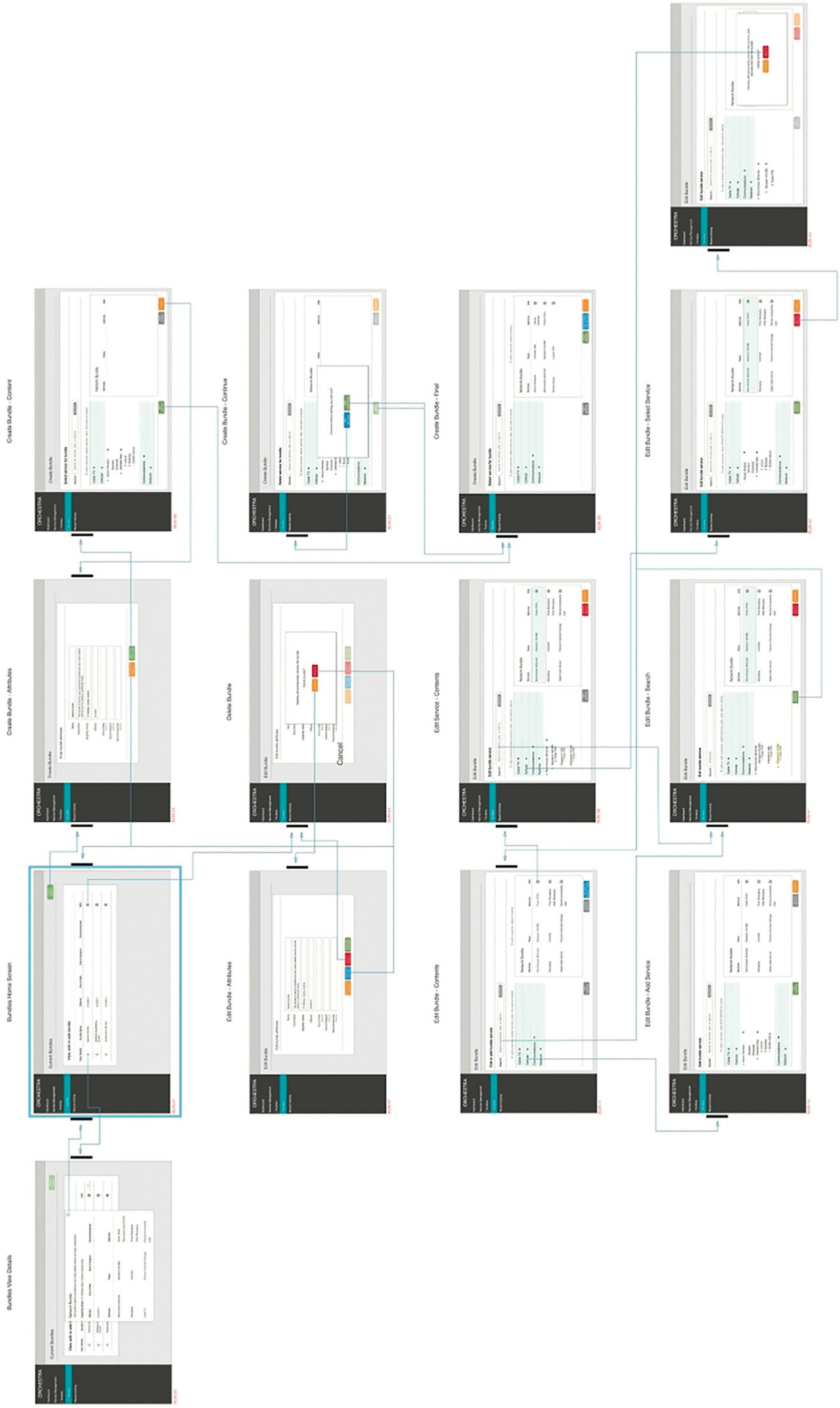

Fig. 18.6
High-fidelity wireframe prototype showing navigation (courtesy of Joe Hutson and Mathew Mathai, Network Infrastructure and Services, Virginia Tech.).

the resources to create a custom design system for your organization, you can adopt popular existing ones, such as Material Design.[2]

It is obvious that reusing templates of design objects from a design system increases your efficiency, but it also can help you with consistency across designs. Tools, such as Figma,[3] allow you to build libraries of such reusable elements, which they call symbols. Symbols can be nested and defined to specify aspects that can be overridden when instantiated. This allows designers, for example, to create a button symbol with a generic text label and to override that label when that symbol is instantiated on the screen. For example, two instances of the same button symbol can have two different labels: “Save changes” and “Discard changes.” Furthermore, updating characteristics of a symbol in the design system will automatically update all instances of that symbol in all wireframe decks that use it. For tutorials on these and other capabilities for improved efficiency, see resources available on Figma’s website. There is an abundance of similar resources and tutorials available just a Google search away.

Once you define the symbols in your design system for your target platform or choose to use one of the popular ones already built and available as a download, you can be very efficient and quick in generating detailed wireframe decks. Even large-scale changes to large wireframe decks can be made rather quickly because of the capabilities of tools, such as Figma.

One additional advantage with using reusable elements from the design system is that you can get from low-fidelity sketches to high-fidelity wireframes very quickly because the elements in the design system are usually specified at the highest end of the presentation fidelity spectrum. Modern design systems even include software code that SE roles could use as-is in implementing the presentation aspects of the UX designs. Therefore wireframes created using a mature design system are pixel perfect from the beginning. The other aspects of fidelity (Section 18.4.1), such as behavior and level of detail in the interaction elements, however, will evolve in time as the team iterates through the designs.

18.4 FIDELITY OF PROTOTYPES

In addition to depth and breadth (Section 18.2), the level of fidelity of a prototype is another characteristic reflecting tradeoffs with respect to completeness achieved and cost/time invested. The fidelity of a prototype reflects how finished it is perceived to be by customers and users (Tullis,

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[2] https://m3.material.io/
[3] https://www.figma.com/

1990). Being finished applies to completeness of content and functionality as well as how refined it is in appearance.

In general, low-fidelity prototypes are less finished but more flexible and can be constructed more rapidly and at less cost. As you progress through stages of development in your project, however, your need for fidelity in prototypes increases. The level of fidelity to aim for depends on your current stage of progress in the project and the purpose for which you plan to use the prototype.

18.4.1 Dimensions of Prototype Fidelity

There are many ways that levels of fidelity in a prototype have been described and used in the past. Here, we focus on the practical role they play in an agile UX process. We can describe the fidelity of a prototype in terms of where it is located within a space having the following dimensions:

- Fidelity of presentation (look and feel or realism)
- Fidelity of behavior (interactivity)

18.4.1.1 Fidelity of presentation

This dimension (fidelity of presentation) is about the realism of the look and feel. How close is the prototype to what the actual product or system will look and feel like? The progression along this dimension is something like this:

- Rough sketches without all elements of the screen shown
- Sketches with most of the details shown but not accurately
- Rough wireframes with most details shown somewhat accurately
- Wireframe specifications with all interaction-element–level details shown as they should be implemented
- Visual comps (pixel-perfect visual representations) as a specification of how the look and feel should be implemented

18.4.1.2 Fidelity of behavior

This dimension (fidelity of behavior) is about how close the prototype is to how the product or system will actually behave. The progression along this dimension is something like this:

- Series of rough sketches, with perhaps arrows showing how the user is envisioned to move from one state of the system to another
- Series of somewhat detailed sketches where an executor moves from one screen to another

- Wireframe deck representing a pre-defined scenario or workflow
- Click-through wireframe prototype
- Fully functional prototype

Certain types of design specifications (e.g., highly interactive patterns with animations and transitions) require this latter type of prototype to express design intent and as a specification.

18.4.2 Low Fidelity Means Initial Cost Effectiveness

Because of the simplicity and obvious lack of fidelity of low-fidelity prototypes, many UX designers overlook their potential. Yet, it is at the lowest end of the fidelity spectrum where you often get the most value from evaluation in terms of user experience gained per unit of effort expended. For finding and fixing most of the obvious usability and UX problems early on, we believe a very low-fidelity wireframe prototype can be enormously effective. A low-fidelity prototype is much less evolved and therefore far less expensive. It can be constructed and iterated in a fraction of the time it takes to produce a good high-fidelity prototype. You can create a design for a set of related user tasks, implement them in a low-fidelity wireframe prototype, evaluate with users, and modify the whole design all within 1 or 2 days.

18.5 BUILD UP PROTOTYPES IN INCREASING LEVELS OF FIDELITY

In this section, we look at the progression through increasing levels of fidelity, as illustrated by starting with low fidelity (especially paper prototypes, such as printouts of static wireframes, for design exploration and early design walk-throughs) and progressing through medium fidelity (such as click-through wireframe prototypes) and high fidelity (programmed functional prototypes), and visual comps for pixel-perfect look and feel.

18.5.1 Very Low-Fidelity Wireframe Sketches to Support Design Idea Exploration in Generative Design

Low-fidelity prototypes are, as the term implies, prototypes that are not faithful representations of the details of look, feel, and behavior, but instead give rather high-level, more abstract impressions of the intended design. The low-fidelity versions usually do not have graphical design elements, such as images, color, or typography. Low-fidelity prototypes are appropriate when design details have not been decided and when they are likely to change.

These quick and rough design representations are almost always hand sketches. They are disposable and short lived, and they evolve rapidly, costing almost nothing to iterate. Therefore these are essentially sketches that come and go as part of design generation (see Chapter 13).

18.5.2 Wireframe Decks Begin to Demonstrate Flow

As you begin to understand the design space and start identifying promising candidates to explore further, and to summarize and clarify it for yourselves in the UX team, you proceed to creating decks of slightly higher fidelity wireframes. They are higher fidelity in that they start representing key interaction elements, such as the broader navigation (but perhaps not all the items that will eventually be shown there).

Using a wireframing tool, draw each wireframe on a slide and group related wireframe slides into ordered wireframe decks. In a demonstration of basic flow, each user action will show a new wireframe as in a slideshow presentation, waiting for the next user action.

Typically, you would show the screens on a laptop displayed for discussion via a projector. Although a little higher fidelity than the design exploration sketches, these low-fidelity wireframes are just static sketches of screens in that there is no real interaction or navigation, so you still have to simulate navigation. You can do this by narrating the action and simulating a potential scenario by pretending to click on interaction widgets on the screen or by saying “the user clicks here” and moving manually to the next wireframe in the deck.

These page sequences can represent the basic flow of user activity within a scenario, but they cannot show all possible navigational paths.

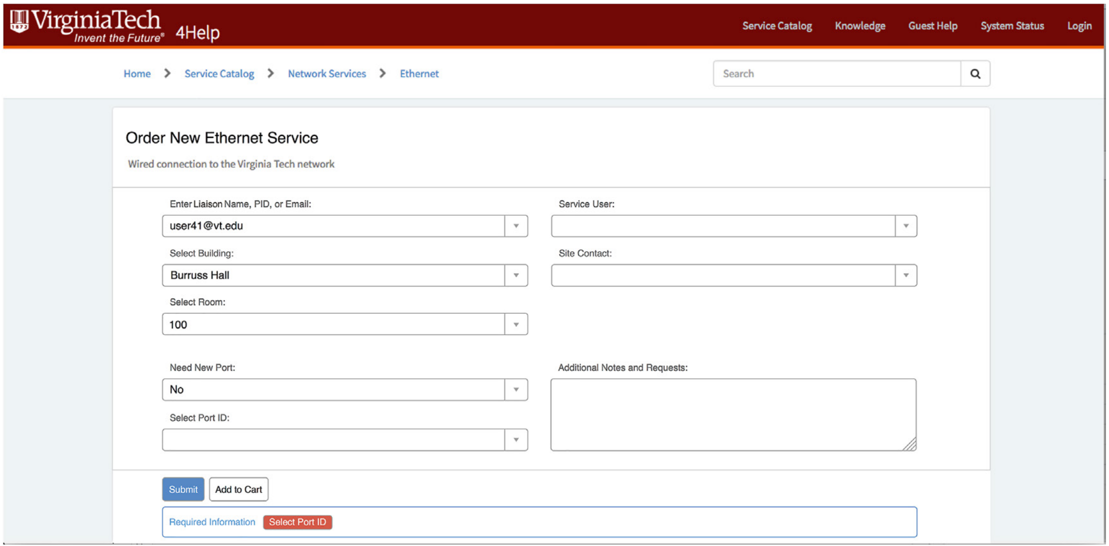
As an alternative, it is sometimes convenient to use paper printouts of the wireframes and move them around on the tabletop to show the design ideas to the UX team and others. During your early design walk-throughs, a prototype executor can manipulate the paper pages to respond to simulated user actions. As you show your stakeholders the prototype on paper, UX team members can write redesign suggestions directly on the paper.

18.5.3 Increased Fidelity Wireframes for Subsequent Design Walk-throughs

As you work with the design and go back to stakeholders with more refined versions, you will want to steadily increase the fidelity of the wireframes, which you can accomplish by adding more screens and more detail to each screen.

18.5.3.1 Example of wireframe sketches for ordering ethernet service with enough fidelity for a design walkthrough

Fig. 18.7 shows a wireframe sketch for a new design to order ethernet service, designed by the Network Infrastructure and Services UX team at Virginia Tech. We used the Sketch wireframing tool, and the broader organization's design system, to incorporate the user interface element and brand styling.



SN_21_HF

Fig. 18.7

Higher fidelity wireframe for ordering ethernet service showing expected colors and style (courtesy of Joe Hutson and Mathew Mathai, Network Infrastructure and Services, Virginia Tech.).

18.5.3.2 Work fast and efficiently

The goal is for the client and users to see some of the UX design very quickly, usually by way of the design walkthrough (see Section 23.3) evaluation technique. Always trade off fidelity (when it is not needed) for efficiency (which is always needed). As an example, if it is not important to have the days and dates on your calendar correspond to the real-world calendar, you can even reuse the calendar grid with the dates already on it.

18.5.4 Medium-Fidelity Click-Through Wireframe Prototypes to Support Early Design Walk-throughs

When you are ready for your earliest design walk-throughs with the client, users, and other stakeholders, you connect these wireframes in a deck by way of some initial hotspots (links or active buttons that can be clicked on to sequence through screens demonstrating interaction flow by simulating navigational behavior). These prototypes usually have no more functionality than that. With this added capability, these wireframe prototypes are called click-through prototypes, which are fast, easy to create, easy to modify, and easily sharable because they are machine readable.

Click-through wireframe prototypes are used for evaluation via design walk-throughs with the UX team and other stakeholders. This is not much more than just moving from one static visual wireframe to another, but because some links now work, the prototype can be operated on a computer by anyone in the group, projecting it onto a screen or monitor for group viewing and discussion. Because the screen changes as you click buttons, it looks like it is really working.

The addition of links takes more effort to create, and even more to maintain as the deck changes, but the added realism is worthwhile. We have had users and clients exclaim that these are so realistic that they look just like the real thing.

18.5.5 Medium- to High-Fidelity Click-Through Prototypes to Support In-Depth UX Evaluation

Once you have the layout of each wireframe settled and have decided on the branding, with all the text, labels, boxes, widgets, and links in place, you may be ready to consider moving on from design walkthrough evaluations to in-depth empirical evaluation (see Chapter 22). This can be with real users who will do the driving and operate the prototype or with experts performing an analytical evaluation.

These days, an in-depth UX evaluation is not something you always have time to do, but if you need to, you should have the right prototype to support it. To support user-based empirical evaluation or expert-based analytic evaluation (see Chapter 23), you will probably need a medium- to high-fidelity programmed prototype with more details of appearance and interaction behavior and possibly even a few connections to system functionality or at least some screens of simulated functionality. HTML5 and CSS3 are common technologies for programmed prototypes and are often developed by a dedicated prototyper or a UX engineer on the UX team.

You can use prototypes containing scripts (written in scripting languages) to give a set of wireframes more ability to respond to user actions, such as links to pieces of real or simulated functionality. This added behavior is limited only by the power of the scripting language, but now you are getting into prototype programming, usually not a cost-effective path to follow very far.

Scripting languages can be relatively easy to learn and use, and being high-level languages they can be used to produce some kinds of behavior (mostly navigational) very rapidly. They are still not effective tools for implementing much real functionality, however.

Preparing for UX evaluation sessions will probably require elaborating all the states of the design relevant to the workflow that is the focus of the evaluation. In projects requiring high rigor and where the stakes for getting it right are high, a high-fidelity prototype, although more expensive and time consuming, can lead to the insight you need; it is still less expensive and faster than building the final product. High-fidelity prototypes can also be useful as advance sales demos for marketing and even as demos for raising venture capital for a company. Aside from these possibilities, high-fidelity prototypes are beyond what you need in most projects, especially in agile development projects.

18.5.5.1 Include decoy user interface objects

If the goal is empirical evaluation with real users, then ensure that the prototypes are representative of the real design, which means they have horizontal coverage. If you include only user interface objects needed to do your initial benchmark tasks (see Section 20.5.1), it may be unrealistically easy for users to do just those tasks. It is more realistic when there are lots of other user interface objects that have to be ruled out when choosing one for a task. The easy way to make this happen is to include other decoy buttons, menu choices, and others, even if they do not do anything, so that participants see more than just the happy path for their benchmark tasks. Your decoy objects should look plausible and should, as much as possible, anticipate other tasks and other paths. When a user clicks on a decoy object, you get to use your "not implemented" message (Section 18.5.5.2). It is also an opportunity to probe users on why they clicked on that object when it is not part of your envisioned task sequence.

18.5.5.2 Make a "this feature not yet implemented" message

"This feature not yet implemented" is the prototype's response to a user action that was not anticipated or that has not yet been included in the design (Fig. 18.8). You will be surprised how often you may use this in UX evaluation with early prototypes.

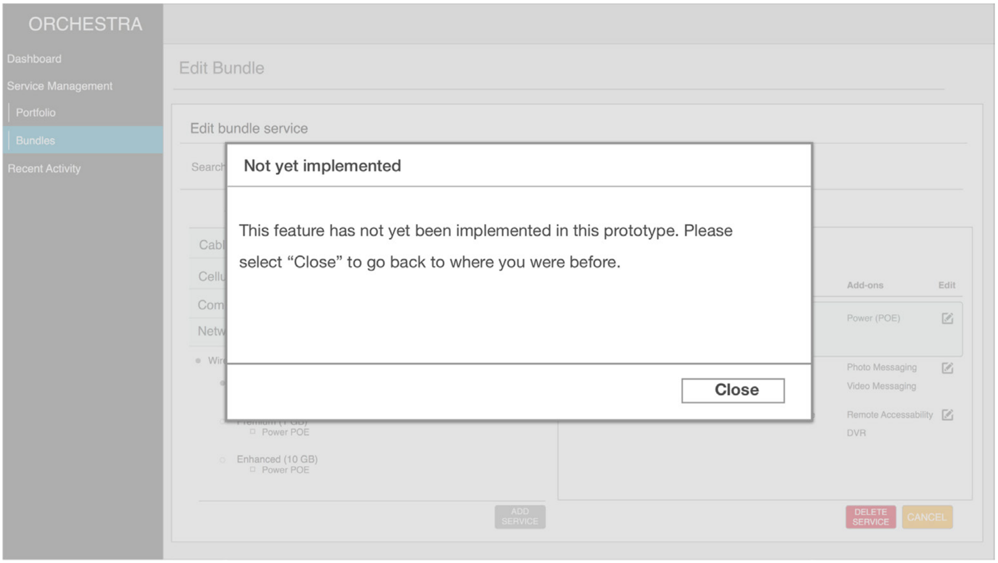


Fig. 18.8

“Not yet implemented” message (courtesy of Joe Hutson and Mathew Mathai, Network Infrastructure and Services, Virginia Tech.).

18.5.6 Medium- to High-Fidelity Prototypes Refined Through Evaluation and Iteration to Hand Off to Software Developers

Finally, after the design ideas are iterated and agreed upon by relevant stakeholders, wireframes (and in rare cases programmed prototypes) can be used as a part of UX design specifications and for design production. Annotate them with details to describe the different states of the design and widgets, including mouse-over states, keyboard inputs, and active focus states. Edge cases and transition effects should now also be described. The goal here is completeness to enable a developer to implement the designs without the need for interpretation.

If the developers do not already have established patterns for color and branding schemes, UX design specifications can be accompanied by high-fidelity visual comps (pixel-perfect application skins) from graphic designers (see Section 11.4.5).

When you hand this prototype off to software developers for implementation, the process requires collaboration across the UX–SE aisle. The UX people own the UX design but not the software that implements it. It is a little like owning a home but not the land it sits on. The hand-off point is a serious nexus between the two lifecycles (see Chapter 26).

Now is the time to go over it with a developer. This prototype:

- Acts as a kind of contract for what is to be built
- Provides the mockup for:
  - Discussing implementation feasibility
  - Checking against software platform constraints
  - Checking for consistency with other features already implemented

Make it clear that you will be back to check the implemented version against this design.

18.5.6.1 The UX team is not yet finished

Because preserving a hard-earned quality user experience in the design is not in the purview of the SE people, the UX people have a strong responsibility to ensure that their UX design comes through with fidelity. This checking of the implementation against the final design is something for which we borrowed the term quality assurance, an essential part of the agile UX lifecycle in the late funnel. If the SE people are the sole interpreters of this process, then there is no way to predict whether the user experience you worked so hard to achieve will still be there in the final product.

Exercise 18.1 Build a Low- to Medium-Fidelity Wireframe Prototype Deck for Your Product or System

Goal: To obtain experience with rapid construction of a low- to medium-fidelity wireframe prototype deck using a modern wireframing tool (e.g., Figma) for some selected user tasks in your product or system (see Section 5.6.1.2).

Activities: This should be one of your most fun exercises, but it can also be quite a bit of work. Make sure that the prototype will support at least the benchmark tasks. (You may have to read ahead in Section 20.5.1 to learn more about benchmark tasks for this.) Add in other decoy interaction design features, widgets, and objects so that the prototype does not look tailored to your benchmark tasks.

Hints and cautions: It is normal for you to do more design work during this exercise to complete details that were not fully designed in previous exercises. Remember, you are learning the process, not creating a perfect design or prototype. Assuming you are doing this as a team, get everyone on your team involved in drawing or using the wireframing tool, not just one or two people. You will be done much faster if everyone contributes. If you are sketching

the wireframes by hand, this is not art class so do not worry too much about straight lines, exact details, and the like. Pilot test to be sure it will support your benchmark tasks for evaluation.
Deliverables: A right, smart, executable wireframe prototype deck that will support your benchmark tasks in UX evaluation.
Schedule: It could take several hours, but it is essential for the exercises that follow.

18.6 SPECIALIZED PROTOTYPES

In addition to the wireframe prototypes that are the bread and butter of today's UX design, for completeness we describe a few other kinds of prototypes to consider in specialized situations.

18.6.1 Physical Mockups for Physical Interactivity

A physical mockup is a tangible, three-dimensional prototype or model of a physical device or product, often one that can be held and often crafted rapidly out of materials at hand and used during exploration and evaluation to at least simulate physical interaction.

If a primary characteristic of a product or system is physicality, such as you have with a handheld device, then an effective prototype will also offer the same kind of physicality in its interaction. Programming new applications on physical devices with real software means complex and lengthy implementation on a challenging hardware and software platform. Physical prototypes are an inexpensive way to afford designers and others insight into the product's look and feel.

Some products are physical in the sense that they are a tangible device that users may hold in their hands. A physical prototype for such products goes beyond screen simulation on a computer, encompassing the whole device. Pering (2002) described an older case study of such an approach for a handheld communicator device that combines the functionality of a personal digital assistant (PDA) and a cellphone. For this kind of product, as an example, it is best to make a prototype from cardboard, wood, or metal that can be handheld in the same way as the product itself would be held.

Designers of the original Palm PDA carried around a block of wood as a physical prototype of the envisioned device. They used it to explore the physical feel and other requirements for such a device and its interaction possibilities (Moggridge, 2007). Fig. 18.9 shows an example of a rough physical mockup of a design for a rickshaw-style cart for transporting people in developing countries.

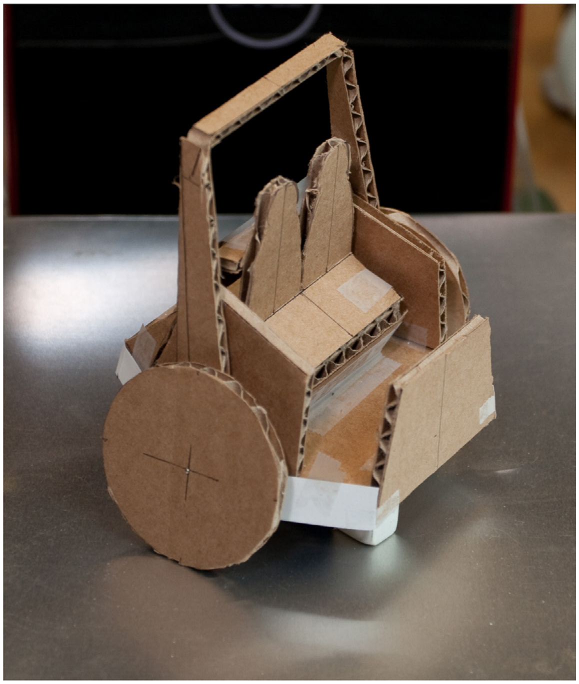


Fig. 18.9

Example of a rough physical mockup of a rickshaw-style cart (courtesy of Akshay Sharma, Virginia Tech Department of Industrial Design.).

A system may also be physical, such as a kiosk. The Ticket Kiosk System is an ideal candidate for physical prototyping. Brainstorm the physical design through ideation and sketches, then build some cardboard mockups that sit on the floor or the ground, add some physical buttons, and have a cutout for the screen about head height. After honing in on the overall look and feel with cardboard, you can make a wooden version to be sturdier, then add physical buttons and attach a touchscreen (e.g., an iPad or a detachable laptop touchscreen) from the inside to fill the cutout and allow some real interaction.

You can use materials at hand and/or craft physical prototypes with realistic hardware. Start off with glued-on shirt buttons and progress to real push-button switches. Scrounge up hardware buttons and other controls that are as close to those in your envisioned design as possible: push buttons, sliders (e.g., from a light dimmer), knobs and dials, a rocker switch, or a joystick from an old Nintendo game.

Even if details are low fidelity, these prototypes are higher fidelity in some ways because they are typically three dimensional, embodied, and tangible. You can touch them and manipulate them physically. If they are small, then you can hold them in your hands. Also, physical prototypes are excellent media for supporting evaluation of emotional impact and other user experience characteristics beyond just usability.

Physical prototyping is now being used for cellphones, consumer electronics, and products beyond interactive electronics, utilizing found objects, junk (paper plates, pipe cleaners, and other playful materials) from the recycle bin, thrift stores, dollar stores, and school supply shops (Frishberg, 2006). Perhaps IDEO[4] is the company most famous for its physical prototyping for product ideation; see their shopping cart project video (ABC News Nightline, 1999) for a good example.

Wright (2005) described the power of a physical mockup that users can see and hold as a real object over just pictures on a screen, however powerful and fancy the screen graphics. Users get a feeling that this is the real product. The kind of embodied user experience projected by this approach can lead to a product that generates user surprise and delight, product praise in the media, and must-have cachet in the market.

Heller and Borchers (2012) created a physical mechanism for individual electric consumers to be visually aware of their levels of electric consumption at any moment as a simple but compelling example of a physical mockup. By integrating a design where the power is consumed (i.e., at the household outlet), they fashioned a display that is there all the time, not something the consumer has to choose to use or must attach. Iterating through a series of software and hardware prototypes, the final step was to add a DIY hardware design printed circuit board behind the outlet, making it a fully functional model. The amount of power being consumed at any moment is displayed very simply by a band of color surrounding the outlet socket—green for low consumption, yellow for medium, and red for high.

See more about physical mockups as embodied sketches (physical embodiments of design ideas) in Section 13.4.3.

18.6.2 Animated Prototypes

Most prototypes are static in that they depend on user interaction to show what they can do. Video animation can bring a prototype to life for concept demos, to visualize new UX designs, and to communicate design ideas. Although animated prototypes are not interactive, they are at least active. An animated prototype is a prototype in which the interaction objects are brought to life via animation, usually in video, to demonstrate dynamically and visually what the interaction looks like.

Löwgren (2004) showed how video animations based on a series of sketches can carry the advantages of low-fidelity prototypes to new dimensions where a static prototype cannot tread. Animated sketches are still rough enough to invite

[4]http://www.ideo.com

engagement and design suggestions, but being more like scenarios or storyboards, animations can convey flow and sequencing better in the context of usage.

Human–computer interaction designers have been using video to bring prototypes to life as early as the 1980s (Vertelney, 1989). A simple approach is to use storyboard frames in a flipbook-style sequence on video or, if you already have a fairly complete low-fidelity prototype, you can film it in motion by making a kind of claymation frame-by-frame video of its parts moving within an interaction task.

18.6.3 Experience Prototyping, the Goal of High-Fidelity Physical Prototyping

As Buchenau and Suri (2000) pointed out, if you are told something, you forget it. If you see something for yourself, you remember it. If you do something for yourself, you understand it.

For some domains, then, for participants to understand the design situation and context well enough to give effective feedback, the prototype they use must allow them to actually do the activity being designed for, to become engaged and immersed in the subjective experience. To appreciate what users will feel, participants need to get beyond being passively exposed to a demo or walkthrough of a prototype and become actively engaged (Buchenau & Suri, 2000).

A very good example is a full-flight simulator for a specific airplane. It is not enough to just look at screens and have someone tell you what would be happening. The pilot in training must experience flight situations in as close a way as possible to the real thing.

Aircraft flight simulators are a special case that can be almost as complex as the aircrafts themselves. Buchenau and Suri (2000) were talking about experience prototypes that succeed in domains not nearly as expensive or as complex. For example, in a project to design for an internet-enabled cardiac telemetry system that included a device to deliver a defibrillating shock to heart patients in the field, participants had to simulate usage of several device designs set in the usage context to give full contextual feedback.

In this case, an experience prototype was used in support of UX research. Participants were given a pager to carry with them on weekends. Getting a message on this device represented a patient getting a rather large electrical shock, with the aim of stopping fibrillation that had been detected by the remote device. They were also given a camera to photograph their immediate surroundings at the time of the proposed shock and a notebook in which to describe the experience, what they were doing, and what it would have been like to be stunned by a real defibrillating shock at that exact moment in time.

Participants quickly understood the necessity for getting a warning before such a shock is administered—if only for safety (in case they were holding a baby or operating a power tool, for example) and to prepare themselves psychologically. Plus, a way was needed to explain the patient's condition to bystanders. In this approach, high fidelity means bringing the participant close to the experience of the real thing.

18.7 SOFTWARE TOOLS FOR MAKING WIREFRAMES

Wireframes can be sketched using any drawing or word processing software package that supports creating and manipulating shapes. While many applications suffice for simple wireframing, we recommend tools designed specifically for this purpose. We use Figma,[5] a wireframing app, to do all the drawing. Figma supports a vast portfolio of plugins that aid designers with many aspects of prototyping, such as enabling click-through prototyping, management of design assets and libraries, and filling in of placeholder textual, numerical, or graphical data into screens.

Beyond this discussion, it is not wise to try to cover software tools for making prototypes in this kind of textbook. The field is changing fast, and whatever we could say here would be out of date by the time you read this. Plus, it would be unfair to the numerous other perfectly good tools that did not get cited. To get the latest on software tools for prototyping, it is better to ask an experienced UX professional or to do your research online.

18.8 SCANDINAVIAN ORIGINS OF PROTOTYPING

Like a large number of other parts of this overall lifecycle process, the origins of prototyping in UX, especially low-fidelity prototyping, go back to the Scandinavian work activity theory research and practice of Ehn, Kyng, and others (Bjerknes, Ehn, & Kyng, 1987; Ehn, 1988), as well as participatory design work (Kyng, 1994). These formative works emphasized the need to foster early and detailed communication with users about design and participation in understanding the requirements for that design.

18.9 SEGUE

We conclude Part 4 with this discussion on prototyping. Next, we get into chapters on evaluating designs in Part 5.

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[5]https://www.figma.com/